How visualization works in Python

PYTHON FOR SPREADSHEET USERS

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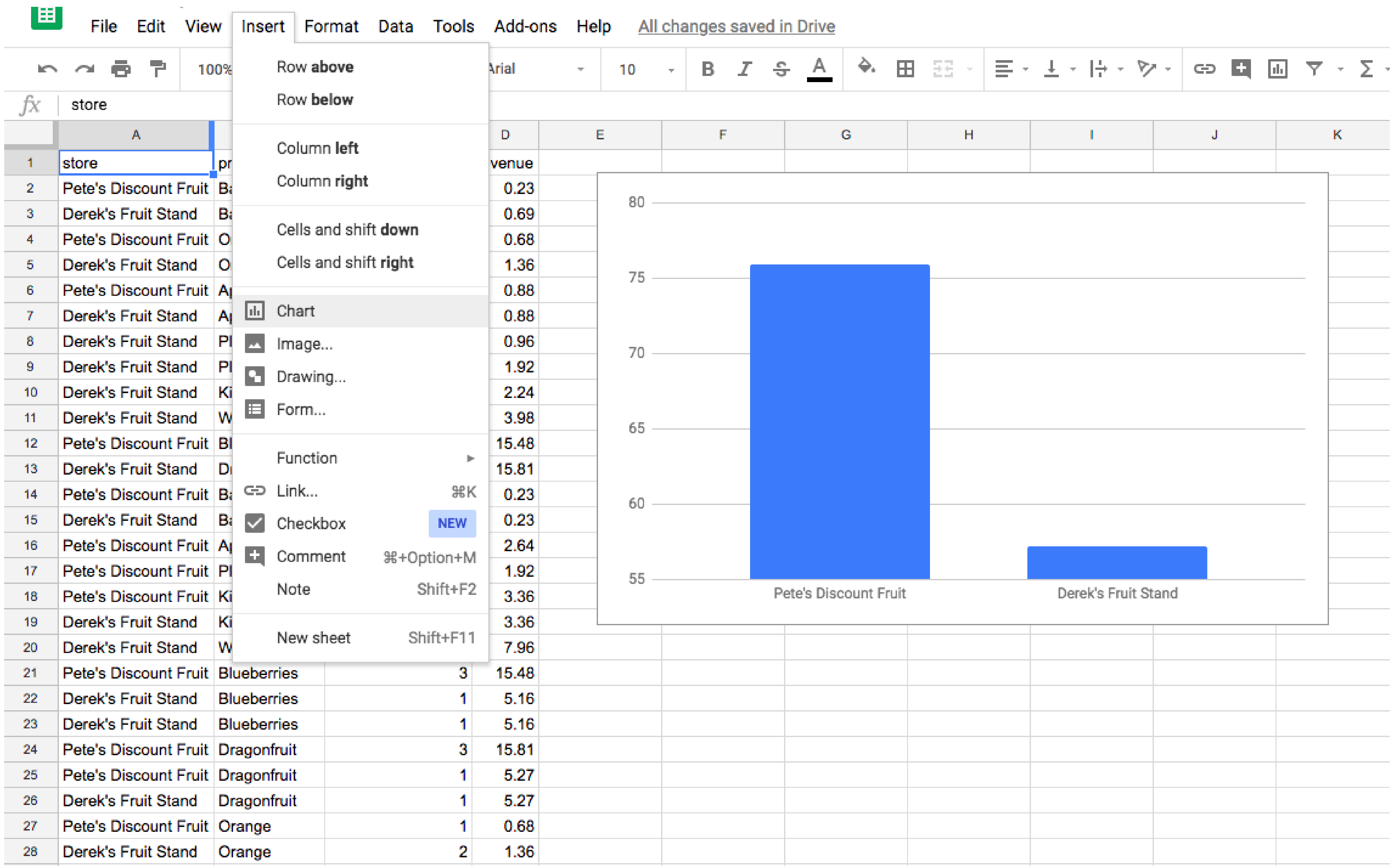
Importing some more packages

```
import seaborn as sns
import matplotlib.pyplot as plt
```

Plotting functions

- `sns.barplot()` - Creates the bar plot
- `plt.show()` - Displays the bar plot

Spreadsheet graphs

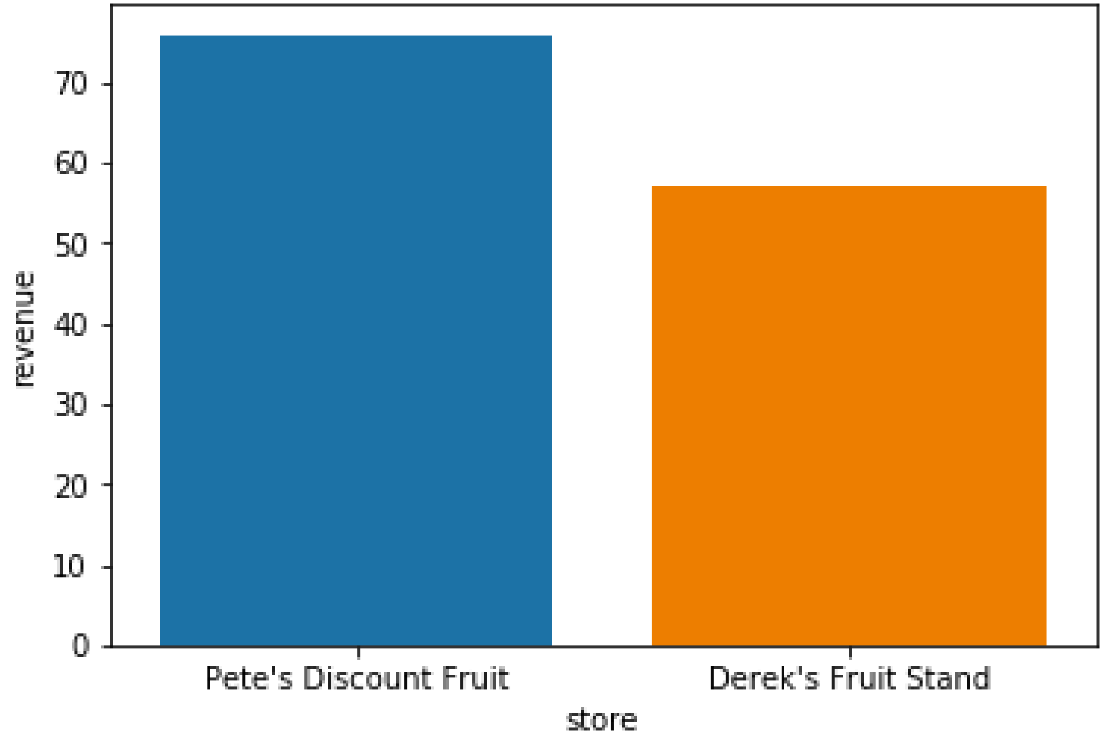


Python graphs

In [3]:
```
totals = fruit_sales.groupby('store', as_index=False).sum().sort_values('revenue', ascending=False).reset_index()
```

In [4]:
```
sns.barplot(x='store', y='revenue', data=totals)

plt.show()
```



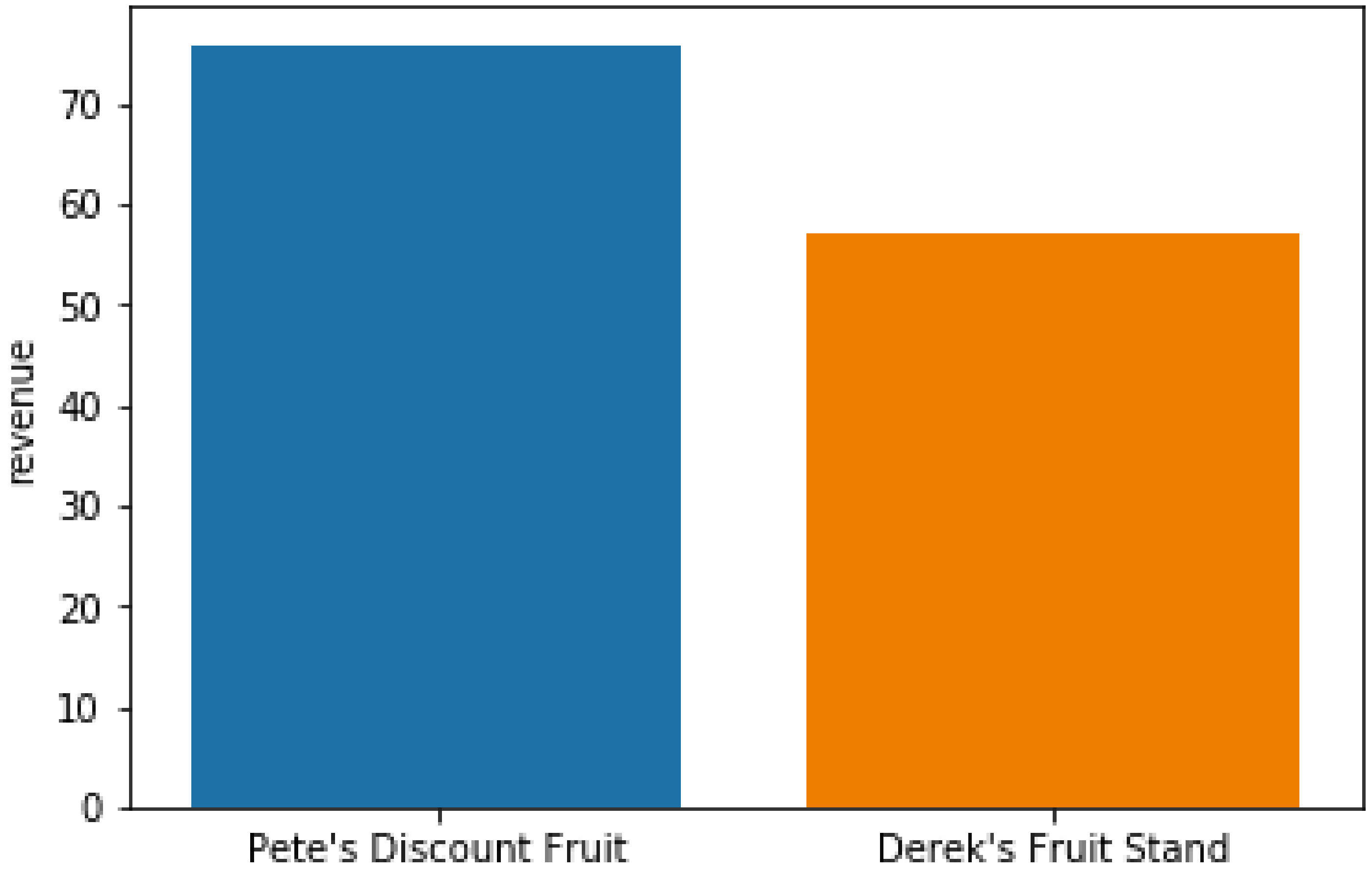
sns.barplot()

```
sns.barplot(x='store', y='revenue', data=totals)
```

plt.show()

```
sns.barplot(x='store', y='revenue', data=totals)

plt.show()
```



plt.savefig()

```
sns.barplot(x='store', y='revenue', data=totals)


plt.savefig('awesome_plot.png')
```

Your turn!

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Building up the barplot

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Previous barplot

```
sns.barplot(x='store', y='revenue', data=totals)


plt.show()
```

Adding labels

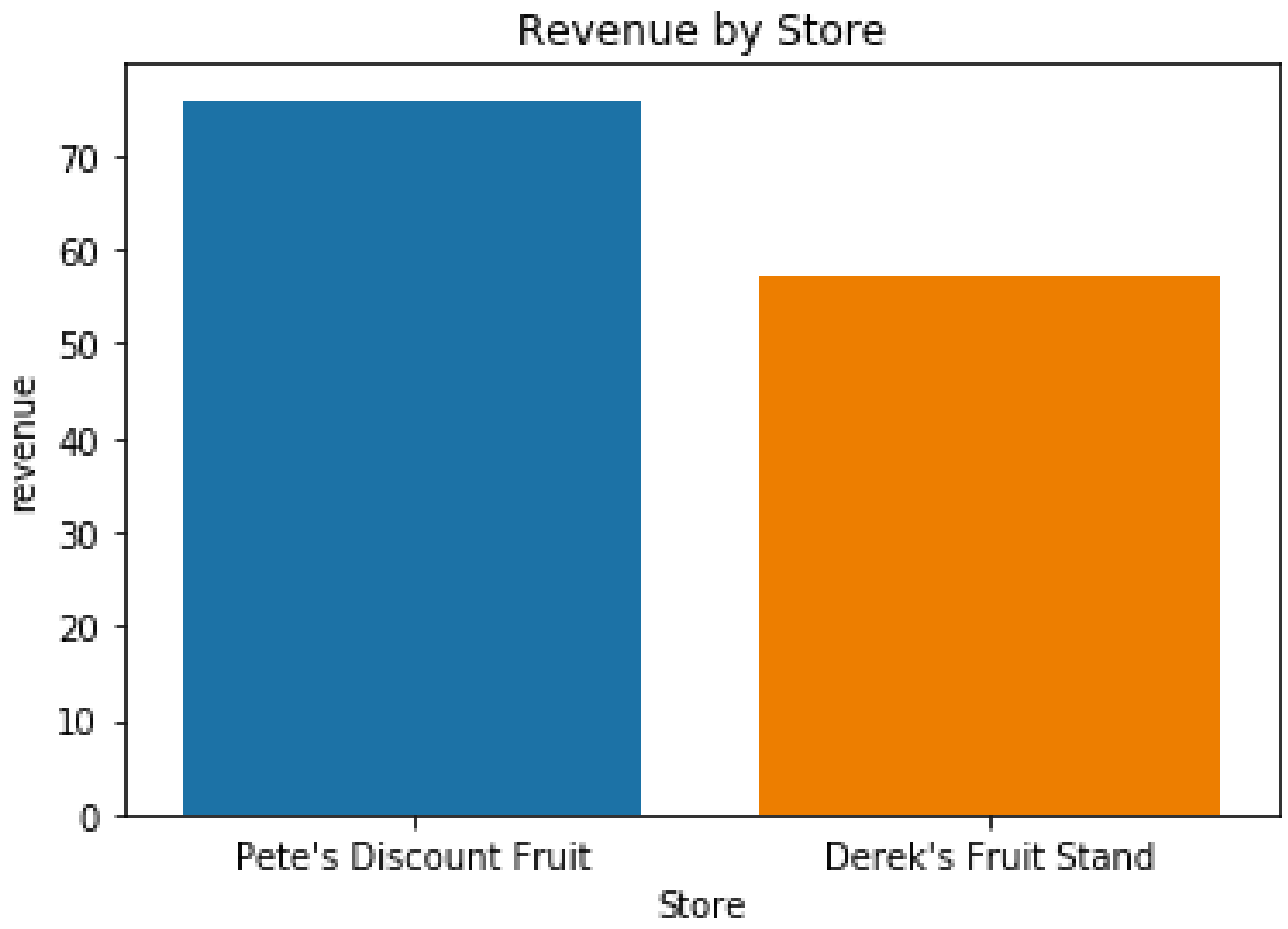
```
sns.barplot(x='store',
            y='revenue',
            data=totals)

# Add a title

# Add x-axis label

# Add y-axis label

plt.show()
```

Adding a title with plt.title()

```
sns.barplot(x='store',
            y='revenue',
            data=totals)

plt.title('Revenue by Store')

# Add x-axis label

# Add y-axis label

plt.show()
```

Adding an x-axis label with plt.xlabel()

```
sns.barplot(x='store',
            y='revenue',
            data=totals)

plt.title('Revenue by Store')

plt.xlabel('Store')

# Add y-axis label

plt.show()
```



Adding an y-axis label with plt.ylabel()

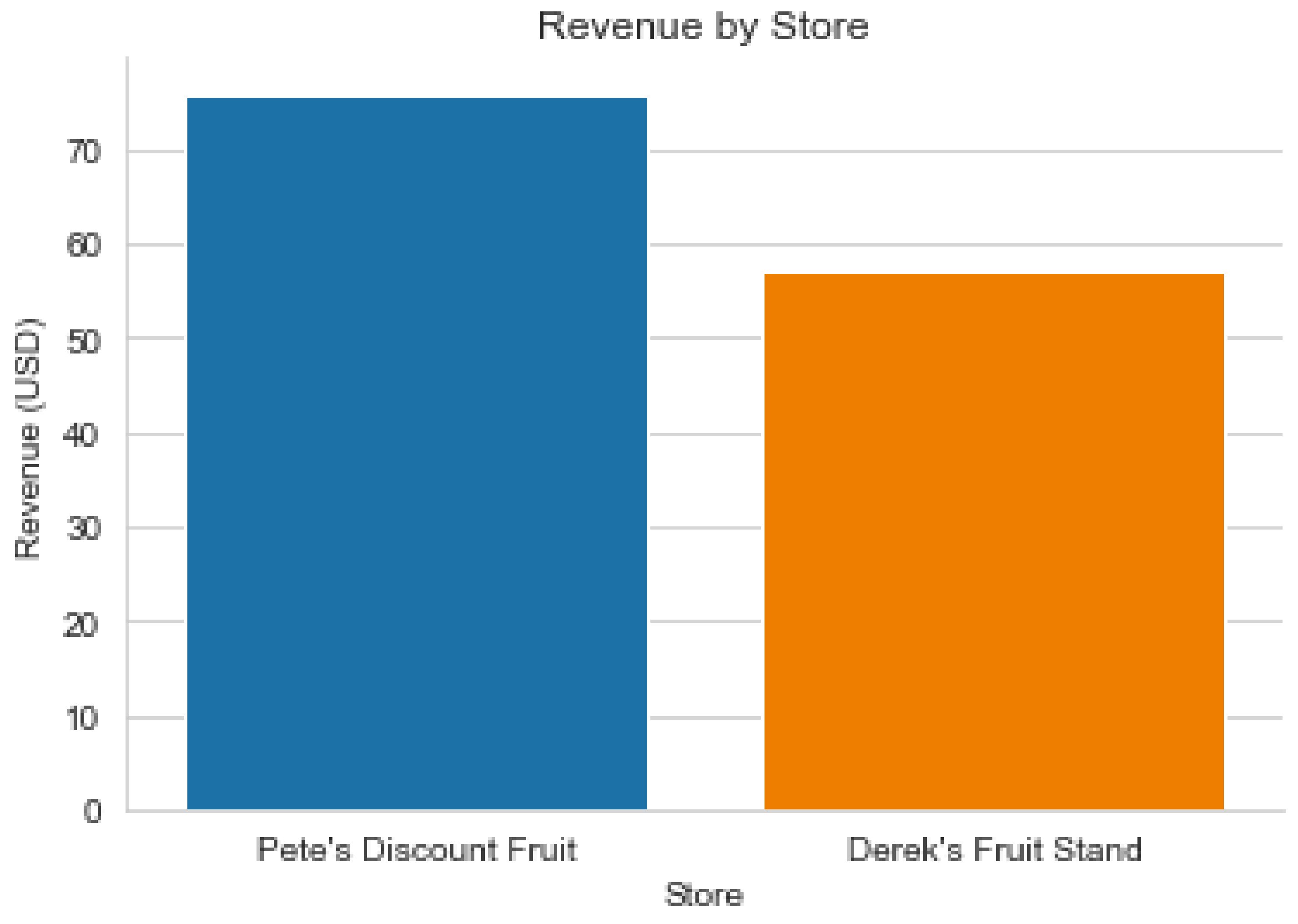
```
sns.barplot(x='store',
            y='revenue',
            data=totals)

plt.title('Revenue by Store')

plt.xlabel('Store')

plt.ylabel('Revenue (USD)')

plt.show()
```

Removing unwanted borders with sns.despine()

```
sns.barplot(x='store',
            y='revenue',
            data=totals)

plt.title('Revenue by Store')

plt.xlabel('Store')

plt.ylabel('Revenue (USD)')

sns.despine()

plt.show()
```

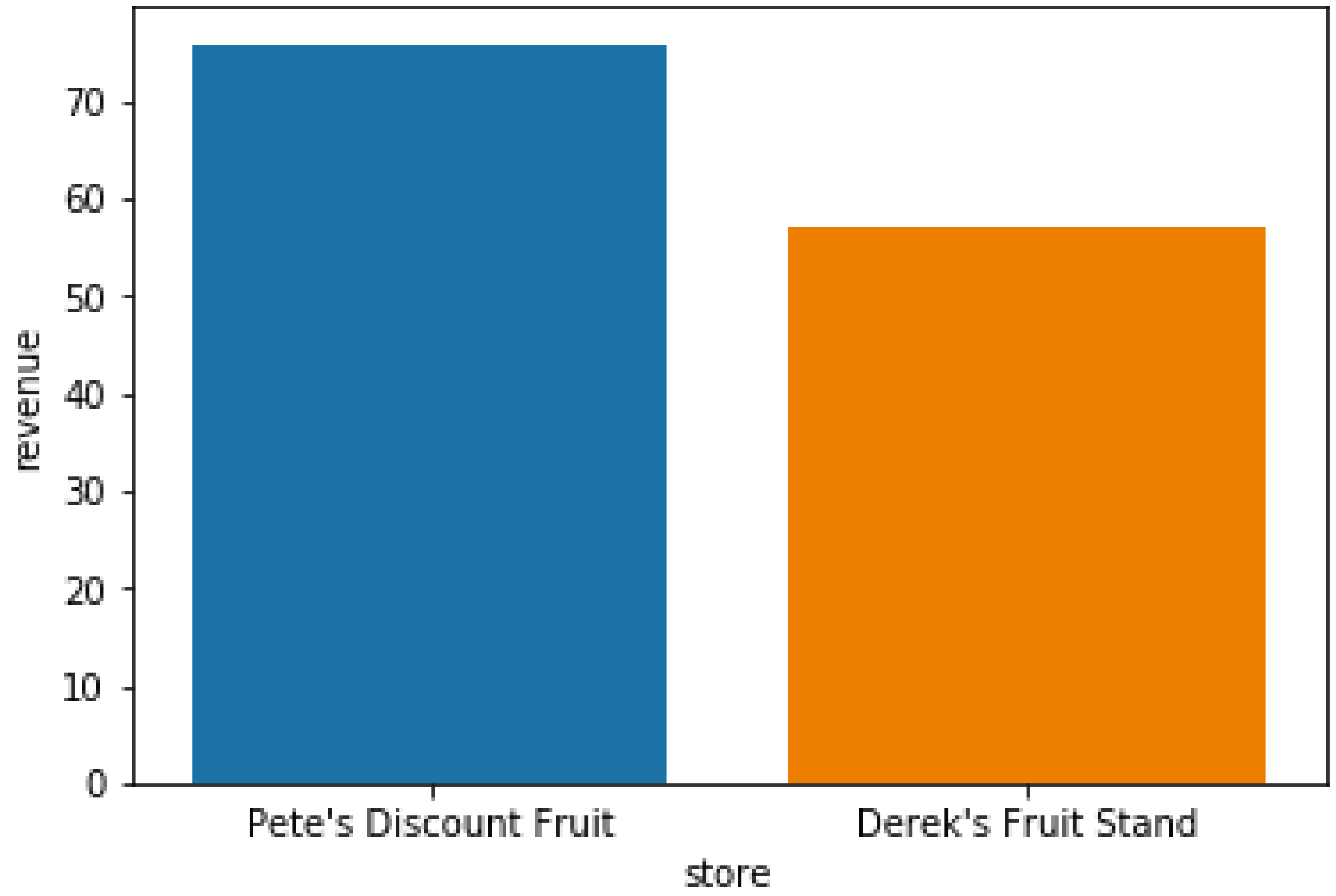
Adding style with sns.set_style()

```
sns.set_style('whitegrid')

sns.barplot(x='store',
            y='revenue',
            data=totals)

plt.title('Revenue by Store')

plt.xlabel('Store')

plt.ylabel('Revenue (USD)')

sns.despine()

plt.show()
```

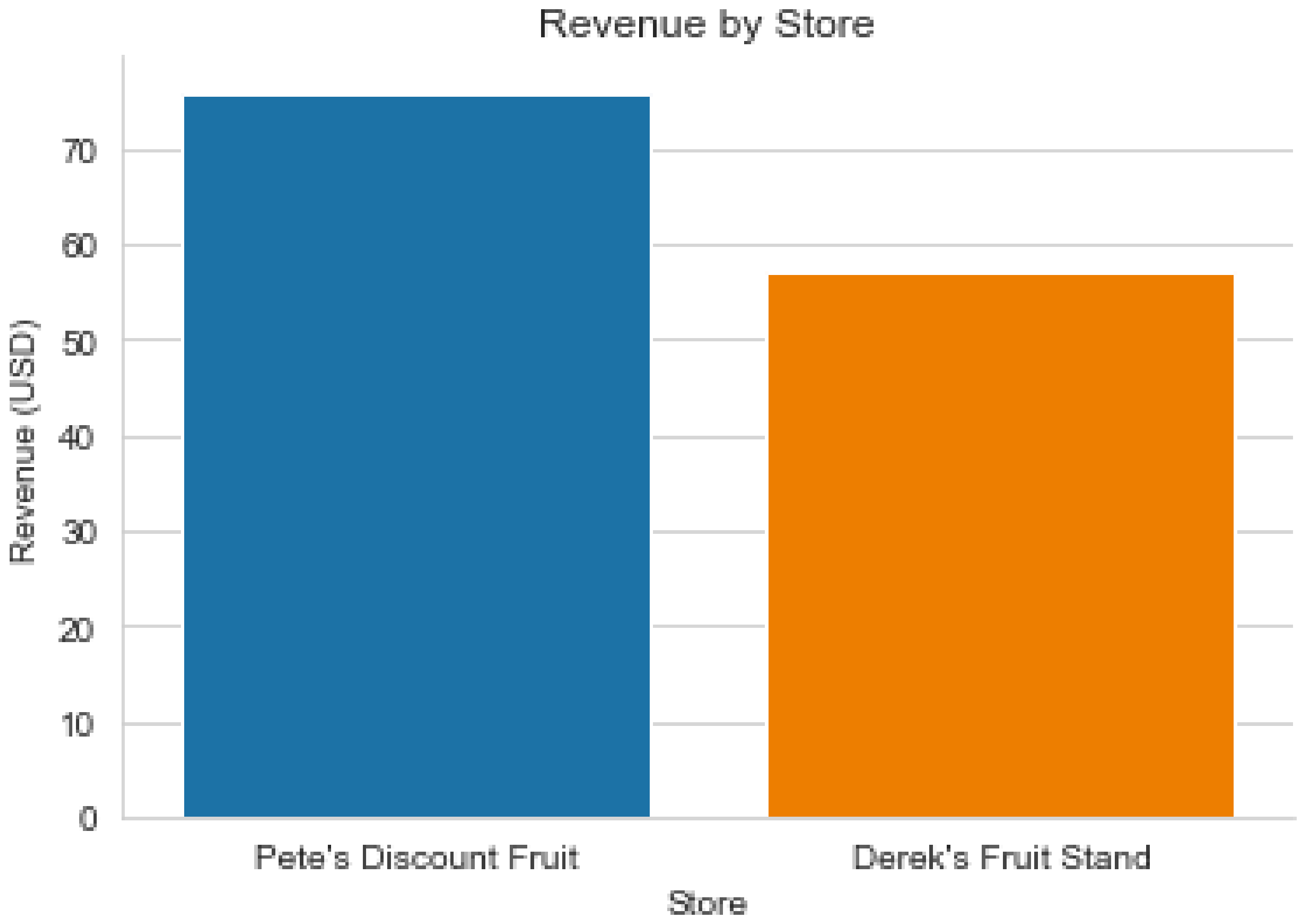


Side by side

BEFORE



AFTER



Your turn!

The power of hue

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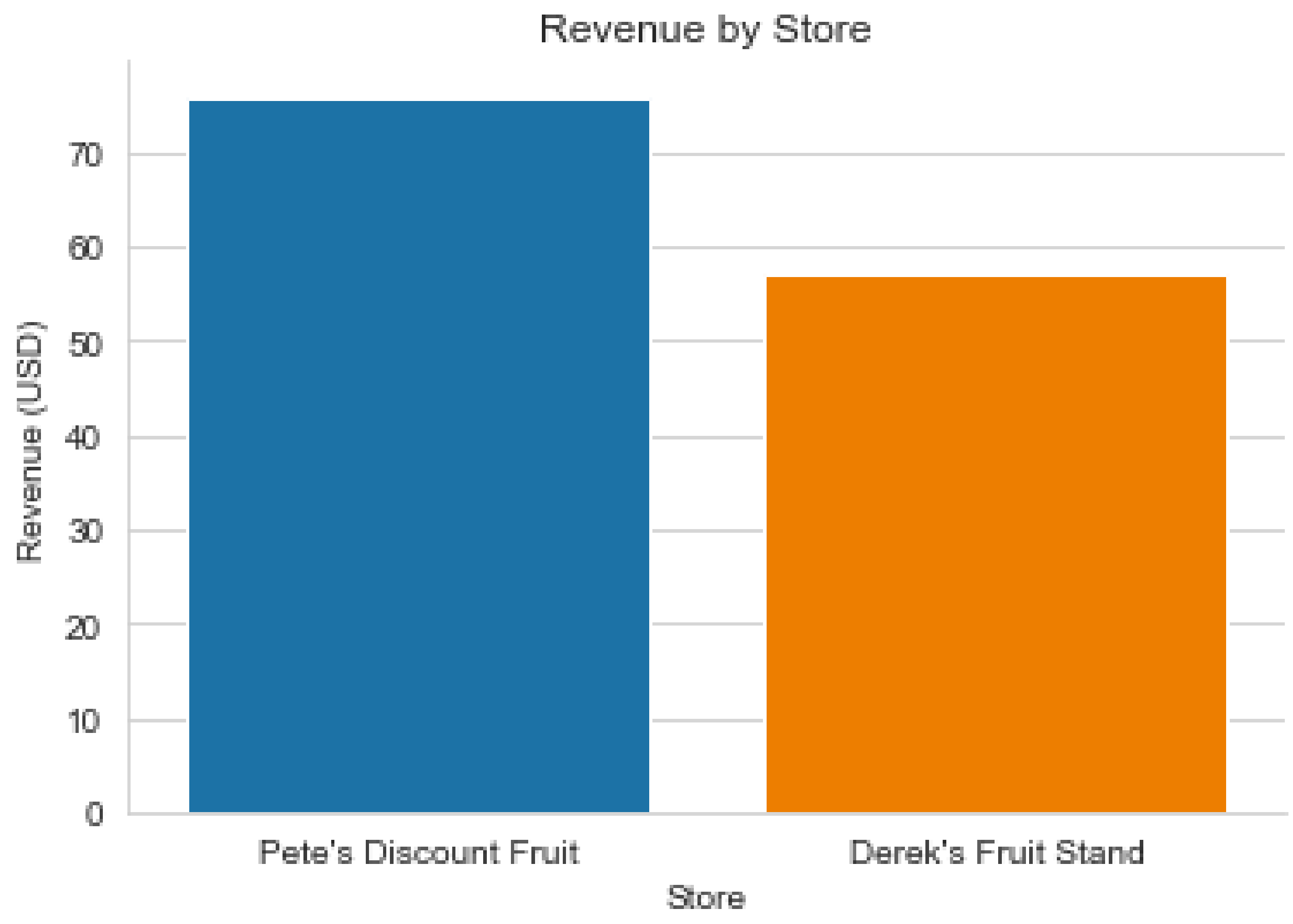
Our nice barplot

THE PLOT



THE DATA BEHIND THE PLOT

```
                   store  quantity_purchased  revenue
0  Pete's Discount Fruit                  30    75.90
1    Derek's Fruit Stand                  27    57.22
```

A more granular barplot

THE DATA BEHIND THE NEW PLOT

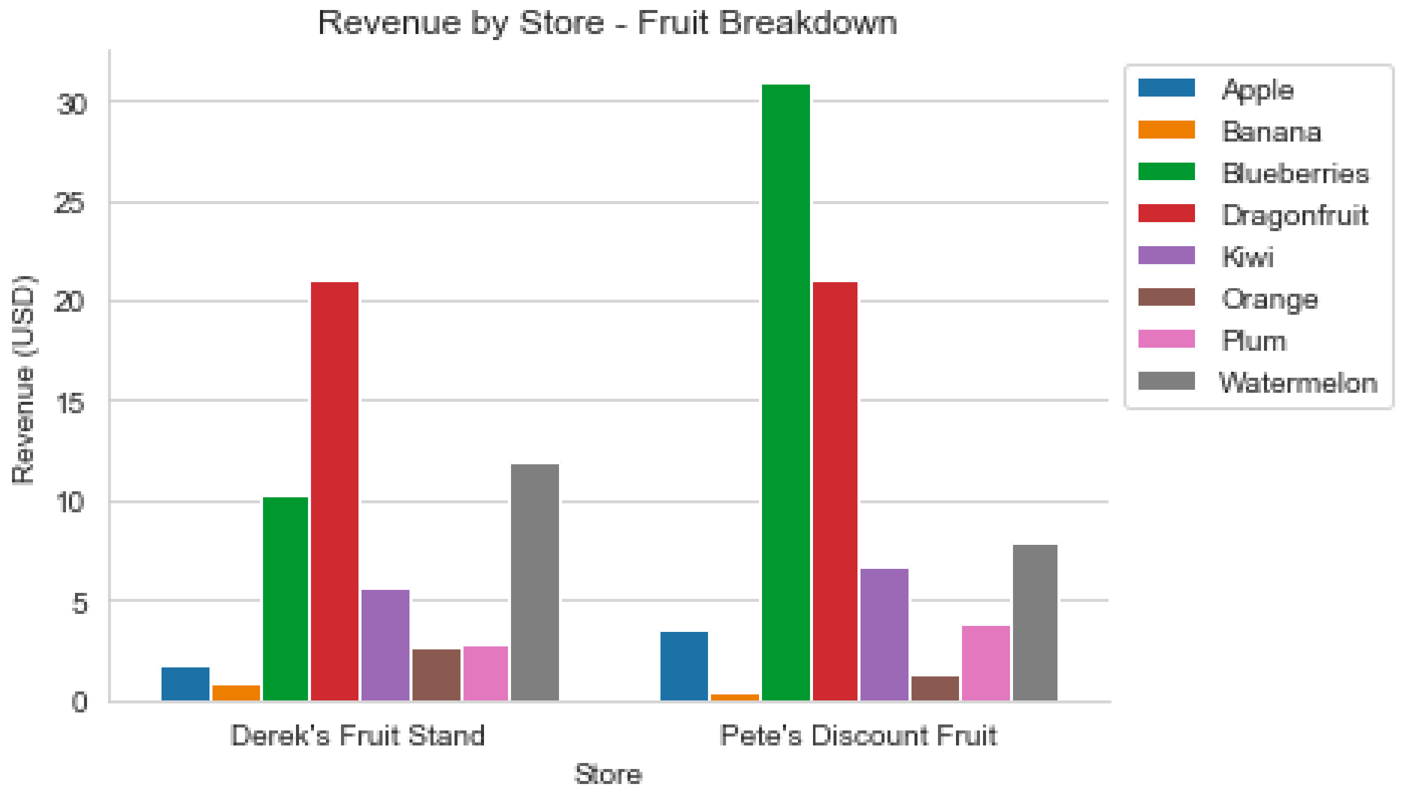
```
                 store product_name  quantity_purchased  revenue
0     Derek's Fruit Stand        Apple                   2     1.76
1     Derek's Fruit Stand       Banana                   4     0.92
2     Derek's Fruit Stand  Blueberries                   2    10.32
3     Derek's Fruit Stand  Dragonfruit                   4    21.08
4     Derek's Fruit Stand         Kiwi                   5     5.60
5     Derek's Fruit Stand       Orange                   4     2.72
6     Derek's Fruit Stand         Plum                   3     2.88
7     Derek's Fruit Stand   Watermelon                   3    11.94
8   Pete's Discount Fruit        Apple                   4     3.52
9   Pete's Discount Fruit       Banana                   2     0.46
10  Pete's Discount Fruit  Blueberries                   6    30.96
11  Pete's Discount Fruit  Dragonfruit                   4    21.08
12  Pete's Discount Fruit         Kiwi                   6     6.72
13  Pete's Discount Fruit       Orange                   2     1.36
14  Pete's Discount Fruit         Plum                   4     3.84
15  Pete's Discount Fruit   Watermelon                   2     7.96
```

A more granular barplot

THE DATA BEHIND THE NEW PLOT

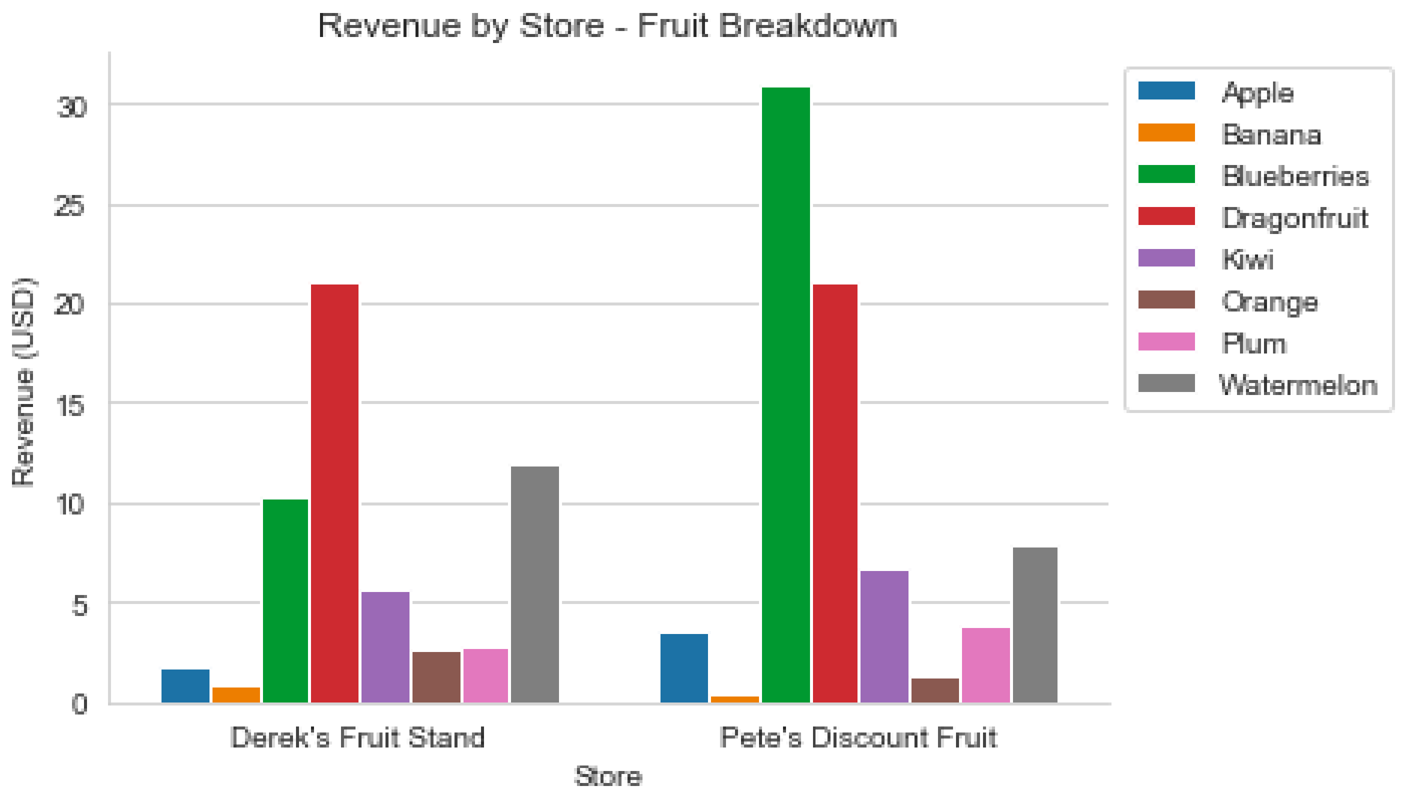
	store	product_name	quantity_purchased	revenue
0	Derek's Fruit Stand	Apple	2	1.76
1	Derek's Fruit Stand	Banana	4	0.92
2	Derek's Fruit Stand	Blueberries	2	10.32
3	Derek's Fruit Stand	Dragonfruit	4	21.08
4	Derek's Fruit Stand	Kiwi	5	5.60
5	Derek's Fruit Stand	Orange	4	2.72
6	Derek's Fruit Stand	Plum	3	2.88
7	Derek's Fruit Stand	Watermelon	3	11.94
8	Pete's Discount Fruit	Apple	4	3.52
9	Pete's Discount Fruit	Banana	2	0.46
10	Pete's Discount Fruit	Blueberries	6	30.96
11	Pete's Discount Fruit	Dragonfruit	4	21.08
12	Pete's Discount Fruit	Kiwi	6	6.72
13	Pete's Discount Fruit	Orange	2	1.36
14	Pete's Discount Fruit	Plum	4	3.84
15	Pete's Discount Fruit	Watermelon	2	7.96

THE NEW PLOT



Hue

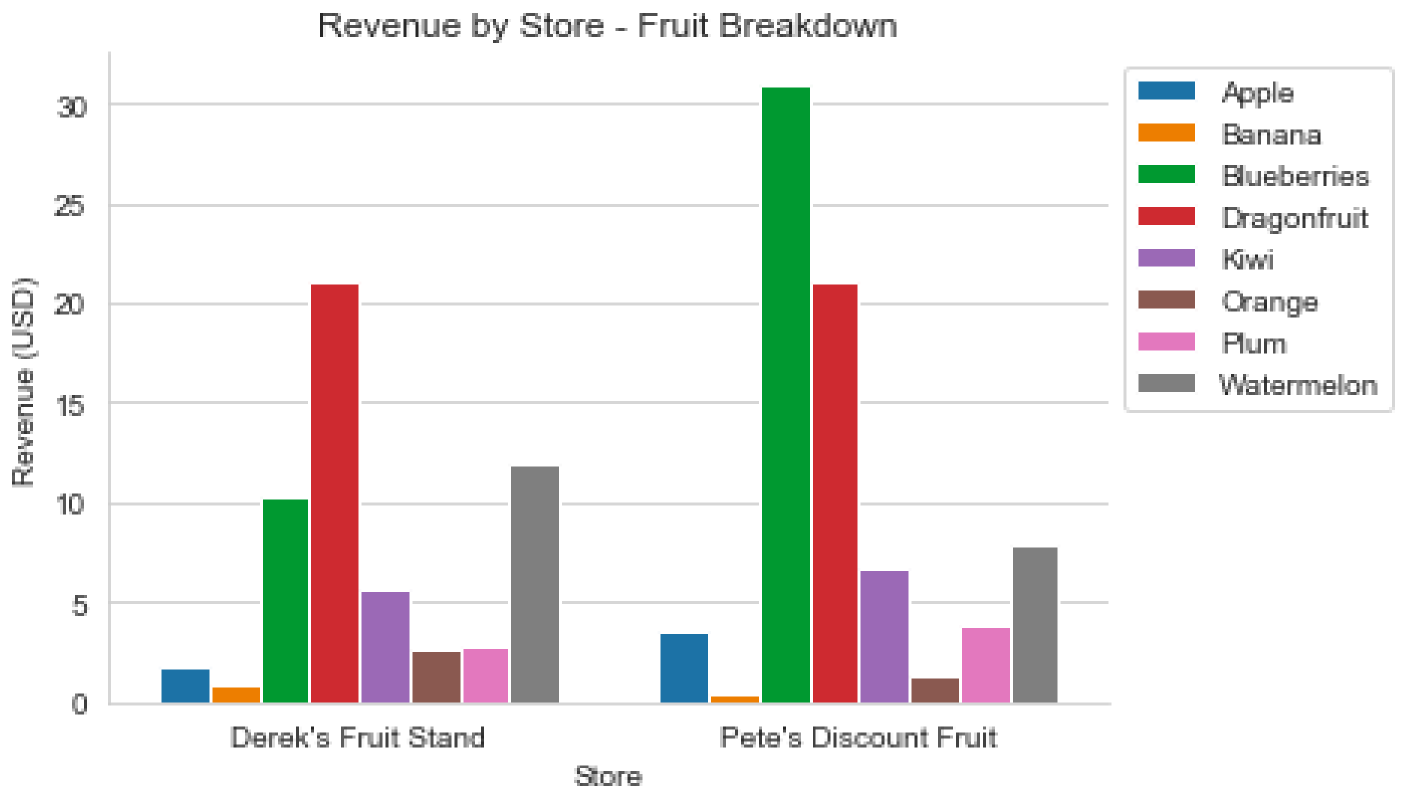
```
sns.barplot(x='store',
            y='revenue',
            data=fruit_store_summary,
            hue='product_name')
```



plt.legend()

```
sns.barplot(x='store',
            y='revenue',
            data=fruit_store_summary,
            hue='product_name')

plt.legend(bbox_to_anchor=(1, 1))
```



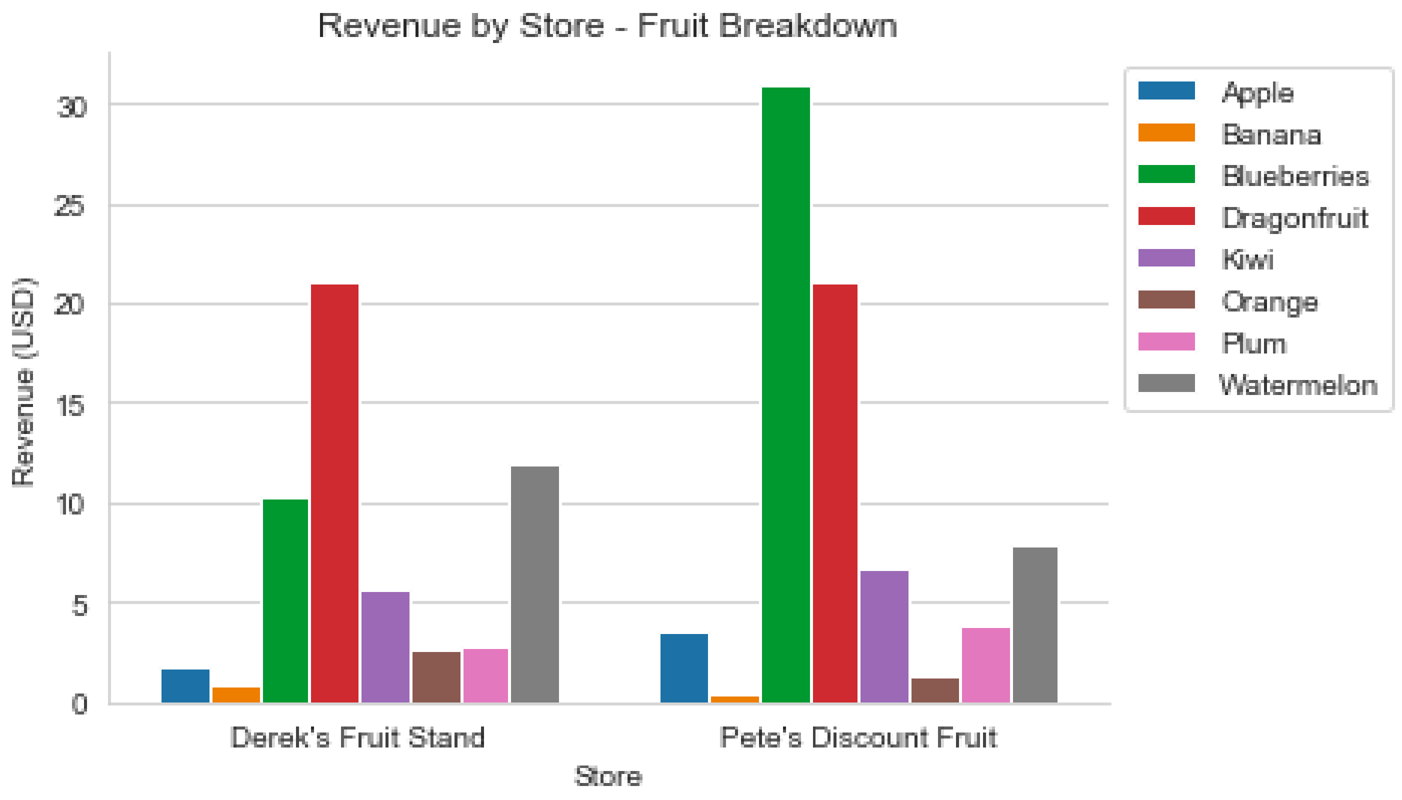
All together

```
sns.set_style('whitegrid')

sns.barplot(x='store',
            y='revenue',
            data=fruit_store_summary,
            hue='product_name')

plt.legend(bbox_to_anchor=(1, 1))

plt.title('Revenue by Store - Fruit Breakdown')
plt.xlabel('Store')
plt.ylabel('Revenue (USD)')
sns.despine()
plt.show()
```



Your turn!

PYTHON FOR SPREADSHEET USERS

Wrapping up

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Data Scientist

Chapter 1

- Importing with `pd.read_excel()`
- Understanding with `.head()` , `.describe()` , and `.info()`
- Filtering with `df[df[column] == 'value']`
- Creating with `df[new_column] = df[old_column] + 1`

Chapter 2

- Pivoting with `df.group_by().sum()`

Chapter 3

- Multiple sheets with `pd.ExcelFile()` and `.parse()`
- `.merge()` and left joins
- Functions, methods, and attributes

Chapter 4

- `sns.barplot()` and `plt.show()`
- `plt.title()` , `plt.xlabel()` , and `plt.ylabel()`
- `sns.set_style()` and `sns.despine()`
- `hue`

Next Steps

- Course Suggestion: Data Manipulation with pandas
- Career Track Suggestion: Data Analyst with Python
- DataCamp Community Resource: Cheat Sheets

Good luck!

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